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Co-evolution and Adaptive Management of the Water-Sediment-Ecology-Socioeconomic Nexus under Climate-Socioeconomic Change

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Abstract

Understanding the coupled interactions among water resources, sediment processes, ecological restoration, and socio-economic development is critical for sustainable river basin management under changing environments, particularly in arid and semi-arid regions. The Ningxia reach of the Yellow River Basin represents a strongly human-impacted system in arid/semi-arid area, where water scarcity, intensive regulation, and ecological vulnerability coexist. In this study, we develop an integrated water-sediment-ecology-socioeconomic (WSES) nexus framework based on System Dynamics to investigate long-term co-evolutionary processes and policy trade-offs in this region. The model is calibrated using long-term observational and statistical data from 2010 to 2021, and future simulations are conducted for the period 2022-2050 under combined scenarios. The model explicitly couples hydrological dynamics, sediment transport, ecological restoration measures, and socio-economic development, and is calibrated using long-term observational and statistical data. By combining afforestation and grassland restoration scenarios with SSPs-RCPs climate-socioeconomic pathways, we simulate system responses from 2010 to 2050. Results reveal pronounced nonlinear relationships and trade-offs between ecological benefits and water availability, identify threshold behaviors in water-sediment regulation, and demonstrate that high-emission pathways substantially amplify hydrological and environmental risks. Moderate ecological restoration (9% forest-grassland growth rate) under low-emission development pathways reduces sediment load by approximately 73% compared to the 3% restoration scenario, while maintaining a stable runoff range and supporting sustained economic growth, emerging as a more robust and adaptive strategy for balancing water security, sediment control, and economic growth.

Keywords: Yellow River Basin; Water-Sediment-Ecology-Socioeconomic (WSES) nexus; System Dynamics Modeling; Changing Environments; SDGs

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1. Introduction

Under accelerating climate change and intensifying human activities, river basins worldwide are experiencing unprecedented pressures on water availability, sediment regimes, ecosystem stability, and socio-economic sustainability [1]. These pressures are particularly acute in arid and semi-arid regions, where limited water resources, fragile ecosystems, and rigid development demands interact nonlinearly, often amplifying system vulnerability. In such contexts, water-related challenges can no longer be interpreted solely as hydrological problems, but rather as emergent outcomes of deeply coupled human-natural systems [2, 3]. Understanding and managing these interactions requires an integrated perspective that explicitly considers the co-evolution of water, sediment, ecological processes, and socio-economic dynamics under changing environmental conditions.

The Yellow River Basin represents one of the most complex coupled river systems in China, characterized by strong water-sediment interactions, intensive human regulation, and pronounced ecological sensitivity [4]. Within this basin, the Ningxia reach occupies a unique transitional position between the upper and middle reaches, situated in an arid to semi-arid climatic zone with extreme water scarcity. Long-term dependence on trans-basin water diversion for irrigated agriculture, combined with large-scale engineering regulation, has fundamentally altered natural runoff and sediment transport processes [5, 6]. Meanwhile, increasing demands from agriculture, industry, and ecological restoration have intensified competition among water use sectors, exposing the region to compound risks including sediment imbalance, ecological degradation, and development constraints. As a result, the Ningxia

reach constitutes a typical, under strong human interference WSES (WSES) coupled system, making WSES Nexus serves as a typical paradigm for coupling the human and natural systems in this region[7].

Despite growing attention to coupled human-natural systems in the Yellow River Basin, existing studies have largely focused on individual subsystems or pairwise interactions, such as water-economy constraints [8], ecosystem service assessments [9], or hydrological responses to human water use [10]. While these studies have yielded valuable insights, they often neglect the critical role of sediment processes and fail to capture the nonlinear feedbacks among water availability, sediment transport, ecological restoration, and socio-economic development. Moreover, many analyses remain static or scenario-specific, limiting their ability to explore long-term system evolution and policy trade-offs under concurrent climate change and socio-economic transitions [11]. Consequently, there remains a clear need for an integrated, dynamic modeling framework capable of quantitatively representing the full WSES Nexus and its co-evolutionary behavior under changing environments [12].

The Coupled Human and Natural Systems (CHANS) framework provides a powerful conceptual basis for analyzing complex human-water interactions, and has inspired a wide range of modeling approaches [13], including agent-based models [14], integrated hydrological-economic models [15], and data-driven learning techniques [16]. Recent advances in nexus research have emphasized multi-sector linkages and trade-offs, particularly in water-energy-food systems [17,18]. However, for river basins where sediment dynamics play a fundamental role in shaping

channel stability, ecological functions, and downstream risks, existing nexus frameworks often remain incomplete. Compared with data-intensive or highly parameterized models, System Dynamics (SD) modeling offers distinct advantages in explicitly representing feedback structures, time delays, and policy-driven system responses [19,20], making it particularly suitable for exploring long-term co-evolution and adaptive management strategies within complex WSES systems.

Against this background, in this study, unlike traditional water-energy-food nexus studies that often neglect sediment dynamics, a complex system dynamics model applicable to WSES in arid and semi-arid regions was constructed to simulate the interactive effects of multivariable and multi-feedback complex systems. We operationalize the CHANS framework by developing a WSES Nexus model tailored to the Ningxia reach of the Yellow River Basin, thereby embedding sediment dynamics explicitly into a coupled human-natural system representation. Grounded in the Coupled Human and Natural Systems (CHANS) framework, we develop an integrated System Dynamics model that explicitly couples hydrological processes, sediment transport, ecological restoration, and socio-economic development. By combining afforestation and grassland restoration policy scenarios with climate-socioeconomic pathways under the SSPs-RCPs framework, the model is used to explore long-term system behavior, nonlinear feedbacks, and critical trade-offs among competing objectives. More importantly, this study moves beyond single-scenario or static analyses by explicitly addressing uncertainty, risk boundaries, and adaptive management under climate-society co-evolution. Through scenario-based simulations from 2010 to 2050, we aim to identify robust

regulatory pathways that balance water security, sediment control, ecological resilience, and socio-economic sustainability. The proposed WSES-oriented modeling framework is not limited to the Ningxia case, but provides a transferable approach for integrated nexus analysis and decision support in arid and semi-arid river basins facing intensified climate variability and human pressures.

2. Study Area and Data

2.1 Study Area

The Ningxia Hui Autonomous Region is located in the arid inland region of northwestern China, in the lower section of the upper reaches of the Yellow River, with a total area of approximately 66,400 square kilometers. It serves as a critical transitional zone from the upper to the middle reaches of the Yellow River [21] (Figure 1). The region receives an average annual precipitation of less than 300 mm, experiences extremely high potential evaporation, and suffers from severe local water scarcity. As a result, its socio-economic development heavily relies on trans-boundary water from the Yellow River [22]. Long-term and large-scale Yellow River water diversion for irrigation has shaped the regional oasis agriculture while significantly altering natural water and sediment fluxes and processes. Consequently, the dynamics of sediment deposition and scouring in irrigation districts and river channels have become a core natural constraint on system stability [23, 24]. Thus, the Ningxia reach of the Yellow River Basin, as the focus of this study, essentially constitutes a strongly human-modified composite system under an arid to semi-arid climatic background [25]. The WSES Nexus represents a typical manifestation of this “natural-anthropogenic” composite system in the region.

Driven primarily by national and regional ecological restoration policies-such as the Grain for Green Program, the Scientific Greening Pilot Demonstration Zone Construction Plan (Ningxia, 2023), and the Territorial Spatial Ecological Restoration Plan (2021-2035)-the forest and grassland area in Ningxia has exhibited a significant increasing trend over recent decades. These policy interventions have led to the planting of approximately 0.4 million ha of forests and the restoration of 66,667 ha of degraded grassland by 2025. Concurrently, climatic factors, particularly the spatial gradient of precipitation (ranging from <200 mm in the north to >550 mm in the south), influence vegetation growth conditions and the effectiveness of restoration efforts. However, policy-driven ecological restoration remains the dominant force shaping vegetation dynamics in this region, with natural succession playing a secondary role. In recent years, systematic ecological restoration efforts have gradually increased the area of planted forests and restored grasslands, promoting a shift toward a more stable ecosystem structure [26]. As a nationally important commodity grain base and energy-chemical industry base, Ningxia has developed a typical “agriculture-industry” competitive water-use pattern [27]. High water consumption irrigated agriculture accounts for the vast majority of total water use, while rapidly developing industry brings continuously growing water demand and pollution load pressures [28]. These economic activities directly form the core of the water resources utilization subsystem and the land use and socio-economic subsystem. Simultaneously, humans constantly adjust the water-sediment relationship through water conservancy projects and agricultural technologies, such as water-sediment regulation [29], water-saving irrigation [30], and

forest-grassland development management [31]. These measures further highlight the “strong human intervention” characteristic of the system.

The core systemic contradiction in the Ningxia reach of the Yellow River Basin manifests as the conflict between the limited supply of water and sediment resources and the continuously growing, structurally rigid socio-economic demand. It is further reflected in the tension between the natural water-sediment processes required to maintain river channel health and the needs to ensure regional food security and support ecological restoration. This distinct regional characteristic determines that studying “water”, “sediment”, “ecology”, and “socio-economic” in isolation cannot reveal the inherent operational mechanisms and risks underlying regional sustainable development. Therefore, this study selects Ningxia to conduct research on a coupled water - sediment - ecology - socio-economic model, fully reflecting the complexity of deep coupling and mutual feedback between natural processes and human activities.

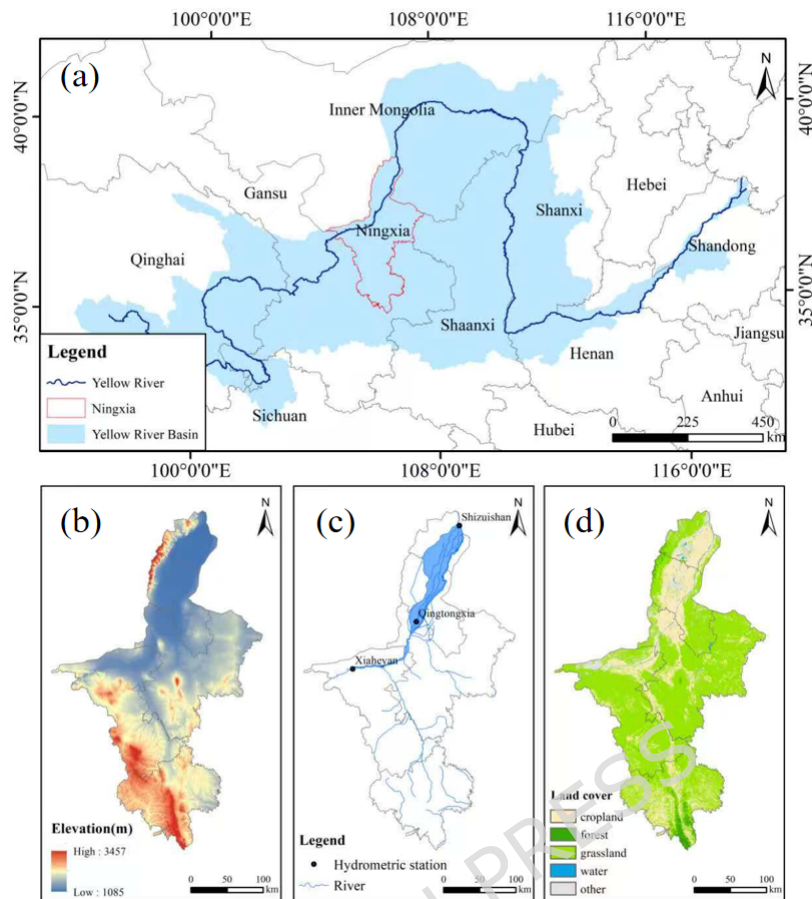


Figure 1. (a) Location of the study area within the Yellow River Basin; (b) Elevation map of Ningxia; (c) River system map of Ningxia; (d) Land use map of Ningxia. These maps were created using QGIS 4.0.1 (<https://qgis.org>) and does not require any permission from anywhere.

2.2 Data Sources

The construction of the complex CHANS model in this study relies on multi-source, long-term data series to ensure an empirical foundation and simulation reliability. Hydrometeorological and sediment data are sourced from long-term observed runoff and sediment load records at major hydrological stations on the main stem of the Yellow River (e.g., Xiaheyan, Qingtongxia, and Shizuishan). These data constitute the core dataset for calibrating the hydrological and sediment subsystems. Precipitation and other meteorological data are obtained from the China Meteorological

Data Service Center (National Meteorological Information Center, 2025). Precipitation data under the four basic RCPs-SSPs pathways are derived from the Coupled Model Intercomparison Project Phase 6 (CMIP6) (World Climate Research Programme, 2025).

Socio-Economic and Water Resource Utilization Data: Key socio-economic and water-use indicators-including population, GDP, industrial structure, cultivated land area, irrigated area, water withdrawal by sector, and water consumption-were obtained from the *Ningxia Statistical Yearbook* (2010-2021) and the *Ningxia Water Resources Bulletin* (2010-2021). These data were used to initialize and calibrate state variables (e.g., total population) and rate variables (e.g., GDP growth rate) within the water-use and socio-economic subsystems.

Policy Parameters: Critical policy parameters, such as water-use efficiency targets delineated in the “14th Five-Year” Plan for Water Security in Ningxia Hui Autonomous Region, the spatial boundaries of ecological protection redlines, and the operational rules of major water-conservancy projects, were incorporated (General Office of the State Council, 2020). For internal system-relationship parameters that are difficult to quantify directly-such as industrial structure ratios and value-added of secondary industry-values were determined through a combination of historical data inversion and references to literature-based estimates [32].

3. Model Development and Scenario Design Methodology

3.1 WSES Nexus Under Changing Environments

Figure 2 illustrates the schematic of the technical framework adopted in this study. Taking a typical region of the Yellow River Basin as a case study, this research analyzes the linkage pathways among system elements and develops a System Dynamics model that captures the mutual feedback between water-sediment resources and socio-economic factors. A coupled analysis of the water-sediment-ecology and socio-economic subsystems is performed. By integrating multi-source data, the complex interactions among these systems are quantitatively investigated, resulting in a complex system model that incorporates water, sediment, ecological, and socio-economic subsystems. Combined with regional ecological management planning and future climate change scenarios, the study elucidates the mutual feedback mechanisms within the basin's water-sediment-ecology-economy system under complex environmental conditions. The findings provide quantitative decision-making support for regional water resource management and sustainable development.

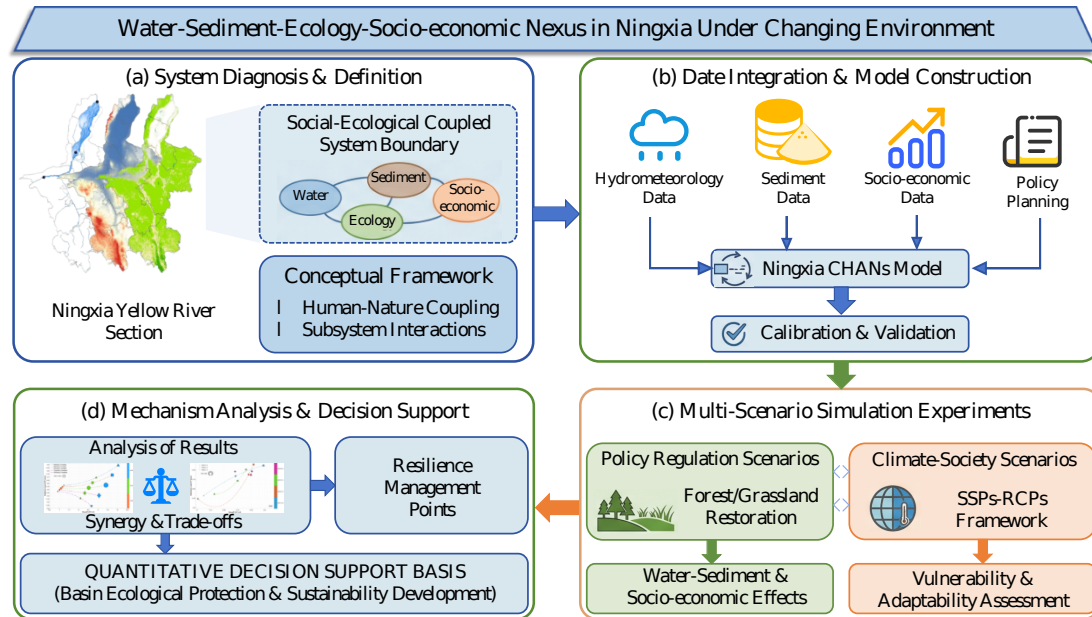


Figure 2. WSES Nexus in the Ningxia region of the Yellow River Basin under changing environments.

3.2 Development of the CHANS Model for the Ningxia Region

3.2.1 Social hydrological driven factors identification

Before determining the model architecture, our study first identified key factors that have been widely validated in Yellow River Basin research (precipitation, land use type, water consumption, urbanization level, etc.) [33]. Using runoff and sediment load as reference sequences, and candidate factors such as precipitation, forest area, grassland area, cultivated land area, water consumption, and urbanization rate as comparison sequences. Grey relational analysis and Deng's relational degree was adopted to quantify the correlation between each candidate factor and the target variable, thereby completing the quantitative screening of factors [34,35].

Let the reference sequence be $x_0(k)$ and the comparison sequence be $x_i(k)$. The grey relational coefficient is calculated using the following formula:

$$x_i(k) = \frac{\min_j \min_k |x_0(k) - x_j(k)| + r \max_j \max_k |x_0(k) - x_j(k)|}{|x_0(k) - x_j(k)| + r \max_j \max_k |x_0(k) - x_j(k)|} \quad (1)$$

Where ρ is the distinguishing coefficient, typically taken as 0.5. The grey relational degree is the average of the grey relational coefficients:

$$g_i = \frac{1}{n} \sum_{k=1}^n x_i(k) \quad (2)$$

3.2.2 CHANS model construction

This study aims to reveal the long-term co-evolutionary trends among water resources, sediment, and the regional economy within the Ningxia reach of the Yellow River Basin, and to identify and assess optimized pathways for achieving regional sustainable development. Spatially, the model focuses on the area along the main stem of the Yellow River in Ningxia, with particular emphasis on the Yellow River diversion irrigation districts and the primary main streams and tributaries. Temporally, it establishes a historical baseline period (2010-2021) for model calibration and validation, and a policy simulation period (2022-2050) for scenario analysis and strategy testing. Figure 3 shows the model framework structure for the Ningxia reach of the Yellow River Basin. In terms of content, the model construction encompasses four core interacting subsystems: water resources, sediment, ecosystem, and socio-economy, which qualitatively describe the WSES Nexus system. Macro-climatic factors, such as annual precipitation, are treated as exogenous variables or stochastic inputs to ensure clarity in the model's internal structure and the operability of the simulations.

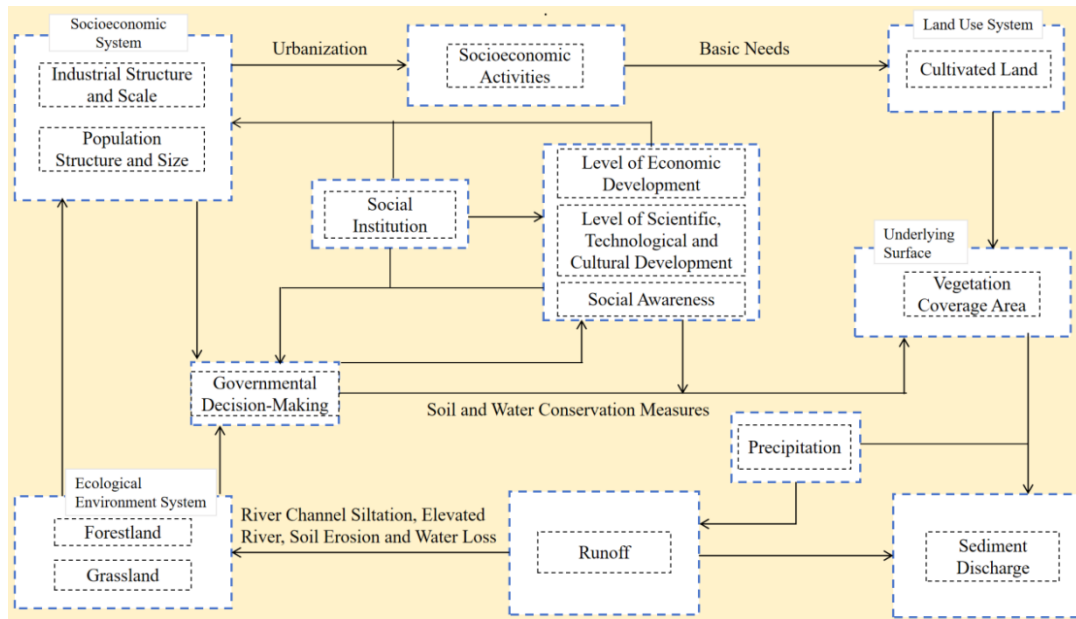


Figure 3. CHANS Model Framework for the Study Area

The model begins with a systematic identification of the four subsystems: water resources, sediment, socio-economics, and the ecological environment. Key state variables, rate variables, and auxiliary variables within each subsystem, along with the direction of their mutual influences, are clearly defined. Building upon this foundation, a Causal Loop Diagram (CLD) for the coupled WSES model system of the study area is constructed (Figure 4). The CLD delineates the causal relationships and feedback pathways among key variables in the system, illustrating how various loops intertwine to form the overall behavioral patterns of the system, thus serving as a crucial bridge connecting qualitative system understanding with quantitative simulation. By closing the causal chains, the research identifies the core feedback loops driving system dynamics, including self-reinforcing positive (reinforcing) feedback loops and self-regulating negative (balancing) feedback loops. Furthermore, the causal structure of key variables in the Ningxia reach of the Yellow River Basin is visualized through Causal Trees for major variables (Figure 5). This process lays the

logical groundwork for subsequent stock-and-flow mapping and equation formulation.

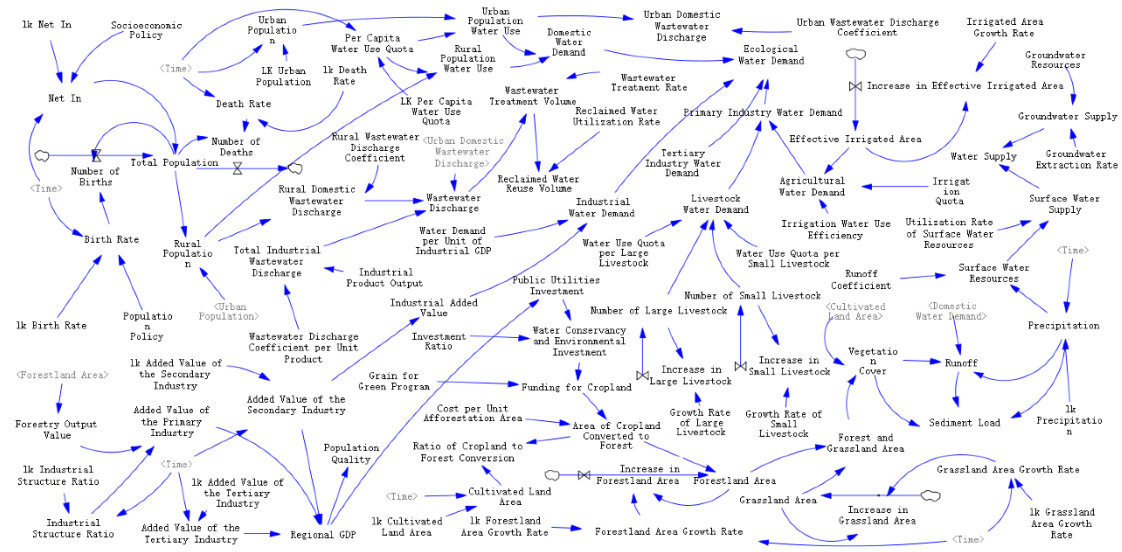


Figure 4. Causal Loop Diagram of the WSES Nexus Model System for the Study Area

Note: The diagram illustrates that the WSES Nexus system comprises social, natural, water resources, and sediment subsystems, including a water resources system centered on runoff, a sediment system focused on sediment load, an ecosystem based on forest and grassland areas, and an economic system measured by GDP.

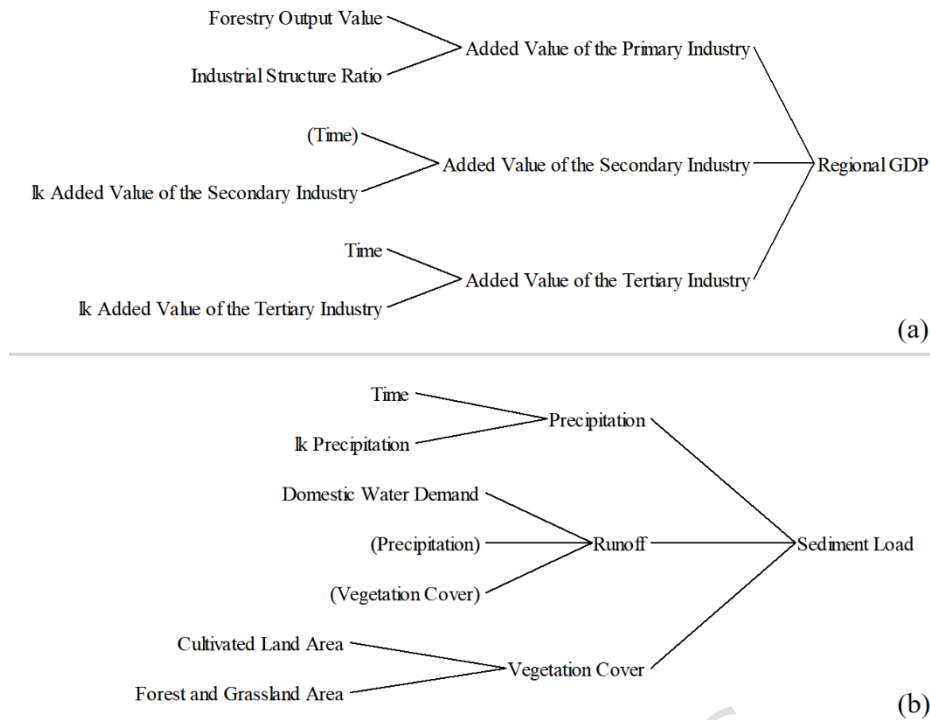


Figure 5. Causal Trees for (a) Regional GDP and (b) Sediment Load

Note: The causal tree illustrates the key driving pathways of regional GDP and sediment load within the system, revealing the multiple causal chains linking economic development and sediment processes.

In the quantitative phase of the model development, the qualitative logic was transformed into a computable simulation model by constructing a variable classification table (Table 1) and defining the corresponding mathematical equations. Building upon the established causal logic, the state variables, rate variables, and governing rules of the system were further clarified, providing a precise framework for system simulation. Specifically, this study first identified and defined the key stocks within the system based on the CLD, which represent the system's accumulated state at any given time. Subsequently, the flows that control the increase or decrease of these stocks, along with their governing functions,

were specified. Concurrently, necessary auxiliary variables and constants were introduced to characterize real-world conversion relationships and policy parameters. On this basis, relevant parameters were determined for all variables using historical data, statistical relationships, and expert knowledge, and mathematical equations were formulated (see Table S1 in the Appendix). This completed the full translation from a conceptual system structure to an executable simulation model.

Table 1. Classification of Variable Types in the System Dynamics Model for the Ningxia Region of the Yellow River Basin

Variable Type	Variable Name
Level Variable (L)	Total Population, Watershed GDP, Grassland Area, Forest Area
Rate Variable (R)	Number of Births, Number of Deaths, Net In-migration Population, GDP Increment, Change in Grassland Area, Newly Added Forest Area
Auxiliary Variable (A)	Runoff, sediment load, Precipitation, Rural Population, Urban Population, Domestic Water Use, Industrial Water Use, Cultivated Land Area
Constant (C)	Investment Ratio, Afforestation Cost per Unit Area, Water Quota for Large Livestock
Table Function (T)	Birth Rate, Death Rate, Urbanization Rate, Per Capita Domestic Water Use, Per Capita Cultivated Land Area in Rural Areas

3.3 Scenario Design under Changing Environments

3.3.1 Scenario development for management policies

The Ningxia reach of the Yellow River Basin is located in the arid to semi-arid region of northwest China, facing long-term ecological constraints such as water scarcity, soil erosion, and land desertification, where the contradiction between development and conservation is prominent[36]. Ecological issues have consistently been a major factor constraining local socio-economic development. Forest and grassland ecosystems play a crucial role in maintaining regional water and sediment balance and improving ecological conditions. However, the current development of forests and grasslands still encounters systemic challenges, including a fragile ecological baseline, prominent degradation issues, and traditional management practices. Therefore, scientifically advancing forest and grassland restoration is an urgent need for alleviating local ecological challenges and regulating the water-sediment relationship in the Yellow River.

The gradient of annual growth rates of forest and grassland area (0.3%–1.2%) is primarily set with reference to the phased forest and grassland growth targets proposed in the Territorial Spatial Ecological Restoration Plan of Ningxia Hui Autonomous Region (2021–2035) and the 14th Five-Year Plan for Forest and Grassland Development in Ningxia, while also taking into account the nonlinear characteristics of cumulative effects in the system dynamics model. Five intensity gradients from low to high are established to characterize the differences in system responses under varying restoration intensities. The specific settings of the main parameters for each scenario scheme are detailed in Table 2.

Among them, S0 is the baseline scenario, assuming that the forest and grassland area remains unchanged over the future period; S1–S5 are scenarios with different forest and grassland restoration intensities, with annual growth rates of 0.3%, 0.6%, 0.9%, 1.2%, and 1.5%, respectively.

Table 2. Key Regulatory Parameter Settings for Each Scenario Scheme

Scenario	Annual Growth Rate of Forest and Grassland Area (%)	Cumulative Growth Multiple (2022-2050)	Cumulative Growth Rate (2022-2050) (%)
S0	0	1	0
S1	0.003	1.039	0.038
S2	0.006	1.183	0.183
S3	0.009	1.287	0.287
S4	0.012	1.400	0.400

3.3.2 Climate change scenario design

The hydrological processes in this region are highly sensitive to climate change, making future climatic uncertainty a primary risk source for long-term planning [37]. To scientifically assess the impacts of future climate change on the WSES system in the Ningxia reach of the Yellow River and to reduce the uncertainty inherent in single-pathway analysis, this study employs a multi-model ensemble analysis approach. Data from multiple global climate models (e.g., BCC-CSM2-MR) under the Coupled Model Intercomparison Project Phase 6 (CMIP6) are selected, covering four Shared Socioeconomic Pathway (SSP) scenarios ranging from low to high carbon emissions (SSP1-2.6 to SSP5-8.5) (Figure 6).

The study defines a historical baseline period (2010-2021) and a future assessment period (2021-2050)

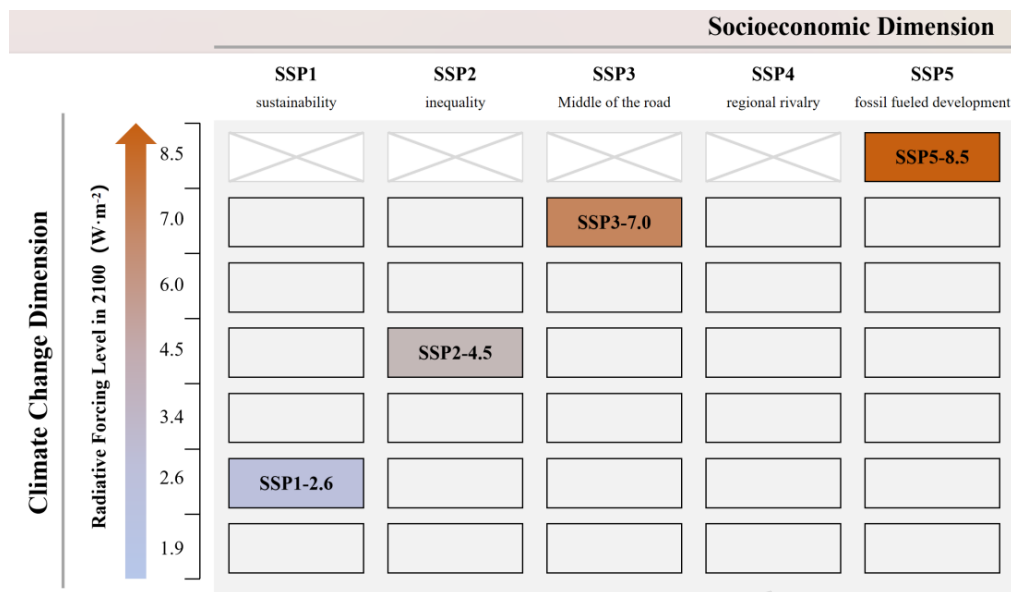


Figure 6. Selected Global Climate Models (GCMs) and the Source Institutions

Note: The selected Global Climate Models (GCMs) and their source institutions are: BCC-CSM2-MR (National Climate Center, China Meteorological Administration) CMCC-CM2-SR5 (Euro-Mediterranean Center on Climate Change, Italy), CAMS-CSM1-0 (Chinese Academy of Meteorological Sciences), ACCESS-CM2 (Commonwealth Scientific and Industrial Research Organisation, Australia), CanESM5 (Canadian Centre for Climate Modelling and Analysis), MIROC6 (Japan Agency for Marine-Earth Science and Technology / The University of Tokyo).

This study selects four SSPs scenario pathways, ranging from low to high carbon emissions, each describing different possible future socio-economic developments in terms of technological levels, water use efficiency, land use, economic growth, and environmental awareness [38-41]. Taking the SSP1-2.6 pathway as an example, Figure 7 presents the precipitation projection data

from different modeling institutions under this scenario (data for other pathways are provided in Appendix Figures S1-S3). Under the low-emission SSP1-2.6 scenario, significant divergence exists among different climate models regarding future precipitation trends. Analysis of the precipitation trend parameters for the models reveals notable inter-model discrepancies. The projected directions of future rainfall change vary across models: MIROC6, CAMS-CSM1-0, and CMCC-CM2-SR5 show positive trends (with slopes of 0.877, 1.652, and 1.107, respectively), whereas CanESM5 and BCC-CSM2-MR project negative trends (with slopes of -1.691 and -0.254, respectively). The intercepts also differ substantially among models (ranging approximately from -3591.730 to -676.970), indicating high uncertainty in the baseline precipitation estimates. Furthermore, the error ranges for the slopes are relatively wide for most models (e.g., slope of 0.877 ± 1.231 for MIROC6), suggesting that the statistical significance of the trends may be insufficient. In contrast, the CMCC-CM2-SR5 model exhibits smaller parameter errors (intercept error ± 1042.160 , slope error ± 0.515), demonstrating higher internal consistency. In summary, precipitation projections under the SSP1-2.6 scenario are significantly influenced by inter-model variability and uncertainty. Therefore, a multi-model ensemble assessment is employed to mitigate bias from any single model and to provide a more robust basis for climate adaptation planning.

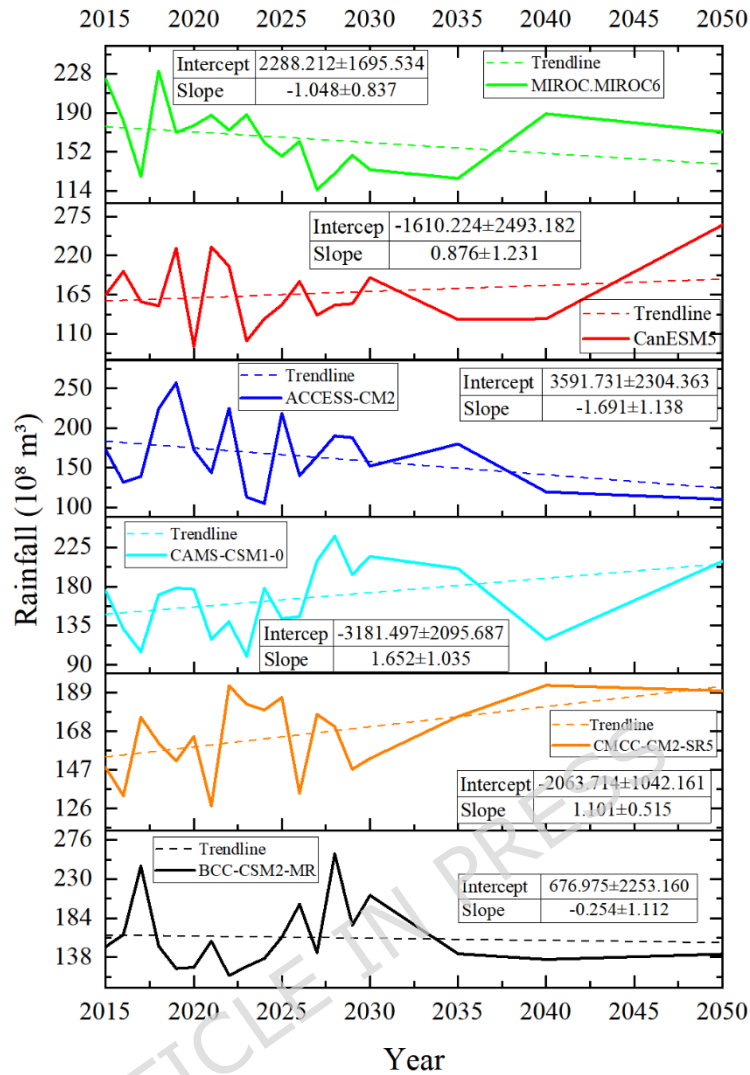


Figure 7. Precipitation Projections from Different Modeling Institutions under the SSP1-2.6 Scenario

4. Results and Discussion

4.1 WSES multi-system simulation model for the Ningxia region

The System Dynamics model constructed in this study has successfully passed systematic calibration tests, including historical data fitting (Figure 8), confirming its capability to reliably simulate the complex mutual feedback relationships within the Yellow River Basin's WSES system. The errors between simulated and historical values for key variables were all within acceptable

ranges. This verifies the structural validity of the model, indicating it has undergone systematic calibration.

Specifically, a comparative analysis of the performance evaluation results for four key variables: sediment load (Figure 8a), runoff (Figure 8b), GDP (Figure 8c), and population (Figure 8d)- reveals the accuracy and challenges in modeling these critical elements within the subsystems of the System Dynamics model. The validation results indicate that the GDP prediction model performs best ($R^2 = 0.977$, NSE = 0.976), with its predicted values closely matching the actual data, it demonstrates that the model has high accuracy in fitting the economic system. In comparison, the population prediction model shows a very high correlation ($R^2 = 0.982$) but a relatively lower Nash-Sutcliffe Efficiency coefficient (NSE = 0.759). For the more complex natural processes, the performance of the runoff simulation ($R^2 = 0.720$, NSE = 0.773) and the sediment load simulation is relatively limited (with both R^2 and NSE around 0.680), highlighting the inherent uncertainty in achieving high-precision simulations within physical systems governed by multifactorial interactions.

Overall, the gradient in model performance (GDP > Population > Runoff > Sediment load) intuitively reflects the varying complexities in modeling and prediction, ranging from socio-economic systems with clearer patterns to natural systems influenced by multiple uncertainties. This finding is consistent with observations in similar System Dynamics studies, where socio-economic variables often achieve higher simulation accuracy than natural hydrological processes [42]. The results also confirm the common observation in carrying capacity and macro-prediction research that the economic subsystem typically exhibits higher

simulation reliability [43,44]. Thus, the calibrated model can serve as a reliable tool for analyzing the synergistic mechanisms within the “Water-Sediment-Ecology-Economy” system and for simulating regulatory policies.

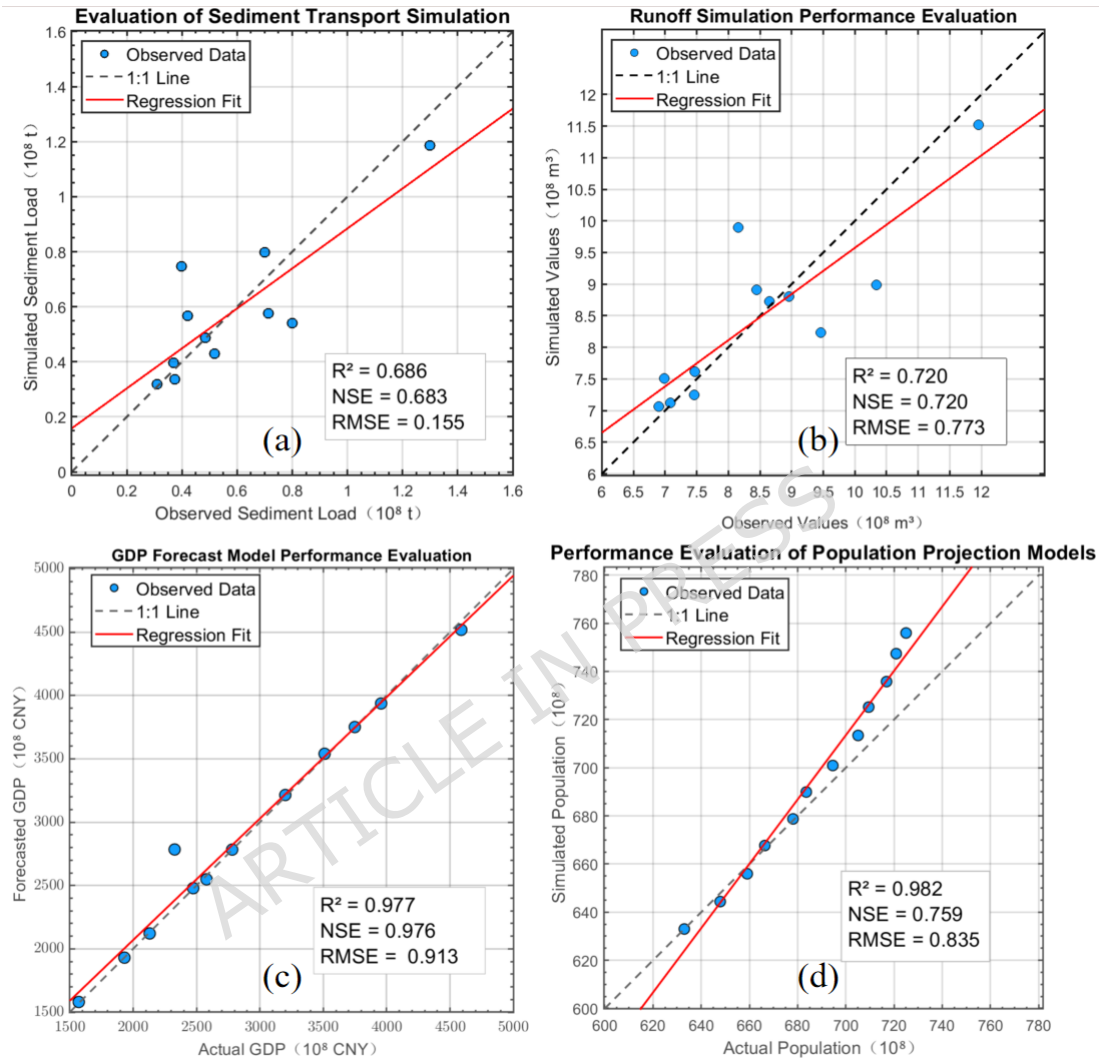


Figure 8. Model calibration against historical data for key variables: (a) sediment load, (b) runoff, (c) GDP, (d) population

4.2 Driven factor identify and sensitivity analysis

The calculated grey relational degrees between each factor and runoff and sediment load are shown in Table 3. The results indicate that precipitation has the highest relational degree with runoff (0.92), followed by forest and grassland area with sediment load (0.92), and water use quota with runoff (0.71), all exceeding the set

threshold of 0.7, indicating that these factors are the core drivers of changes in runoff and sediment. The results showed that the relational degrees between soil and water conservation measures and runoff and sediment were 0.53 and 0.62, respectively, both below the set threshold of 0.7. Meanwhile, the sediment retention effect of check dams was not included in the model due to the lack of long-term, spatially matched quantitative data. Changes in industrial water withdrawal structure accounted for less than 5% of the total water consumption in the Ningxia section, and their contribution was limited; therefore, they were not introduced as independent variables. These screening results are mutually validated by the Mann-Kendall trend test and correlation analysis, providing an objective basis for variable selection in the subsequent system dynamics model.

Table 3. Grey relational degrees of variables

Driving factor	Grey relational degree with runoff	Grey relational degree with sediment load
Precipitation	0.918	0.895
Forest area	0.724	0.782
Grassland area	0.708	0.756
Water consumption (total)	0.753	0.638
Cropland area	0.615	0.672
Urbanization rate	0.562	0.521
GDP	0.589	0.534
Soil and water conservation measure intensity (afforestation area)	0.532	0.617

Change in industrial water withdrawal structure	0.483	0.426
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To further validate the robustness of the model, this study selected three key parameters-annual growth rate of forest and grassland area, precipitation, and water use efficiency-and conducted sensitivity analysis using the one-variable perturbation method (perturbation amplitude of $\pm 10\%$). Table 4 presents the sensitivity test results. The results indicate that the sensitivity coefficient of precipitation to runoff is ($S = 1.21$), and to sediment load is ($S = 1.35$), both exceeding 1, classifying precipitation as a highly sensitive parameter. This indicates that precipitation is the primary natural driver of changes in runoff and sediment in the Ningxia section. The sensitivity coefficient of forest and grassland area growth rate to sediment load is ($S = -0.78$), whereas its influence on runoff is relatively small ($S = -0.23$). The sensitivity coefficient of water use efficiency to runoff is ($S = -0.34$), while its effect on sediment load is minor ($S = -0.18$). The variations in key output variables (runoff, sediment load, GDP) all fall within acceptable ranges, demonstrating that the model structure has good robustness.

Table 4. Sensitivity coefficients of parameters

Parameter	Sensitivity coefficient for runoff	Sensitivity coefficient for sediment load	Sensitivity level
Precipitation	1.21	1.35	Highly sensitive
Forest and grassland area growth rate	-0.23	-0.78	Moderately sensitive (to sediment load)

Water use efficiency	-0.34	-0.18	Low sensitivity (to runoff)
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4.3 WSES Nexus driven by regional management policies

To assess the long-term effects of ecological restoration policies on the comprehensive management of the Ningxia reach of the Yellow River, this study, based on regional planning documents such as the *Regional Land Space Ecological Restoration Plan (2021-2035)*, designed five policy scenarios (S1-S5) with varying intensities of afforestation and grassland area expansion, comparing them against a baseline scenario (S0). The simulation period spans from 2022 to 2050. This study selected SSP3-7.0 as the climate path for driving policy scenario simulations, serving as an intermediate path for radiative forcing values. This approach is suitable for evaluating the robustness of ecological restoration policies under moderate climate stress conditions, ensuring conservative and rational assessment of the WSES relationship in the context of climate change. The analysis delves into the chain reactions-cascading effects on water sediment processes and socio-economic development in the Ningxia reach-induced by changes in forest and grassland coverage under different growth rates.

Based on the validated System Dynamics model, the long-term impacts (2010-2050) of different forest and grassland restoration scenarios (with annual area growth rates ranging from 0.3% to 1.5%) on the sediment processes in the Ningxia reach of the Yellow River Basin were simulated (Figure 9a), alongside a box plot of the sediment load differences between scenarios S0 and S5 (Figure 9b). Over the 28-year cumulative growth period, even seemingly

small annual growth rates resulted in significant medium to long term cumulative effects due to compound growth. The increase in vegetation coverage varied noticeably from scenario S1 to S5. The results indicate that, under all policy scenarios, the changes in sediment load in the river basin exhibit distinct phased characteristics: sediment load continued to decrease from 2010 to 2030; however, during 2031-2050, when the forest and grassland growth rate fell below a 9% threshold, the sediment load began to increase again. Exceeding this threshold led to a continued decline, with the sediment reduction effect strengthening significantly as the restoration intensity increased. High growth scenarios (e.g., S5 with an annual growth rate of 1.5%) substantially reduced both the soil erosion modulus and the sediment delivered into the Yellow River, which is of great significance for alleviating channel siltation. This study quantitatively reveals, from a subsystem perspective, the ecological feedback mechanism between forest and grassland vegetation and sediment load, providing a scientific basis for formulating watershed soil conservation and ecological restoration strategies.

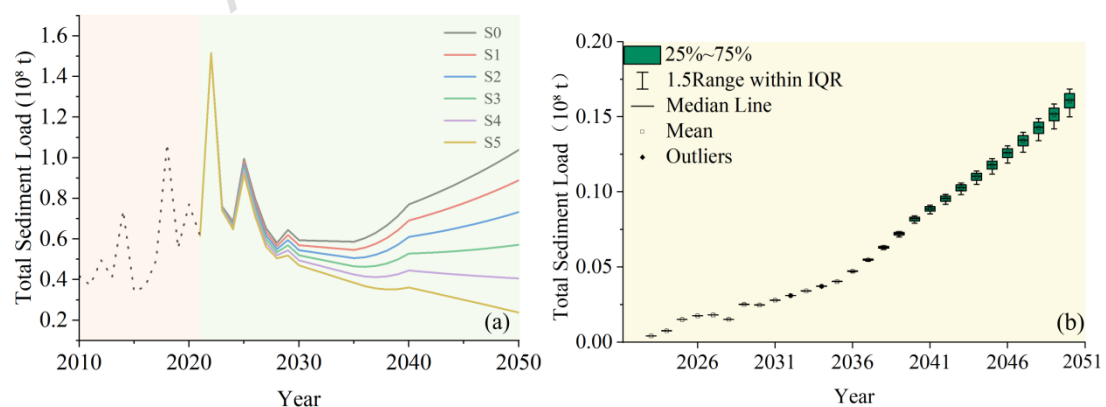


Figure 9. (a) Historical measurements and future simulations of sediment load under scenarios S1-S5; (b) Box plot of sediment load differences between scenarios S1-S5

Figure 10 illustrates the long-term evolution trends of runoff in the Yellow River Basin under different forest and grassland restoration scenarios (Figure 10a) and the corresponding box plot of runoff differences between scenarios S0 and S5 (Figure 10b). Under different forest and grassland restoration scenarios, runoff exhibits phased changes: a decrease from 2022 to 2028, followed by a recovery from 2029 to 2050. The initial decline in runoff is attributed to a reduction in annual precipitation (from 18.468 billion m^3 to 16.479 billion m^3) and enhanced vegetation interception, which weakens runoff generation capacity. In the later period, the recovery in precipitation (from 16.091 billion m^3 to 17.479 billion m^3) drives the restoration of runoff. This evolution reveals the nonlinear relationship and temporal lag effects between vegetation water consumption and hydrological regulation in arid regions, indicating that forest and grassland restoration exerts a dual regulation function on the water-sediment system, thereby providing a quantitative basis for coordinating ecological protection and water resource management.

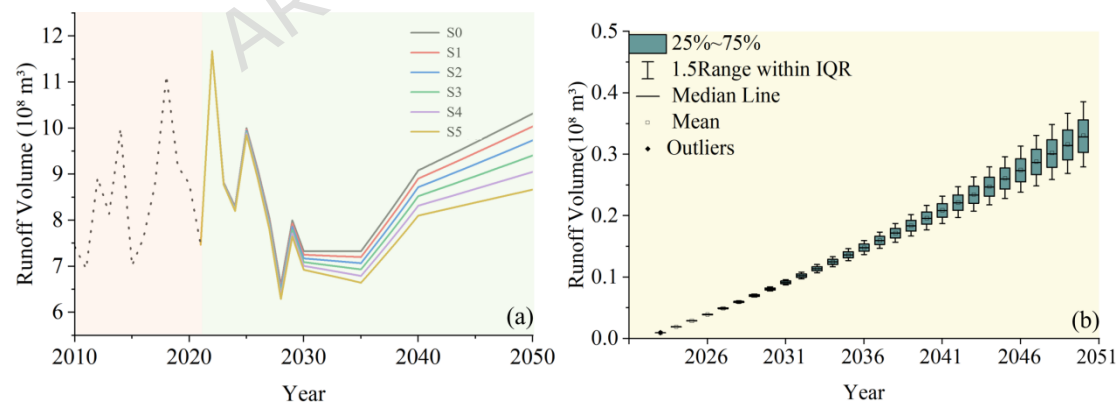


Figure 10. (a) Historical measurements and future simulations of runoff under scenarios S1-S5; (b) Box plot of runoff differences between scenarios S1-S5

Figure 11 illustrates the long-term evolutionary paths of GDP in the Yellow River Basin under different forest and grassland

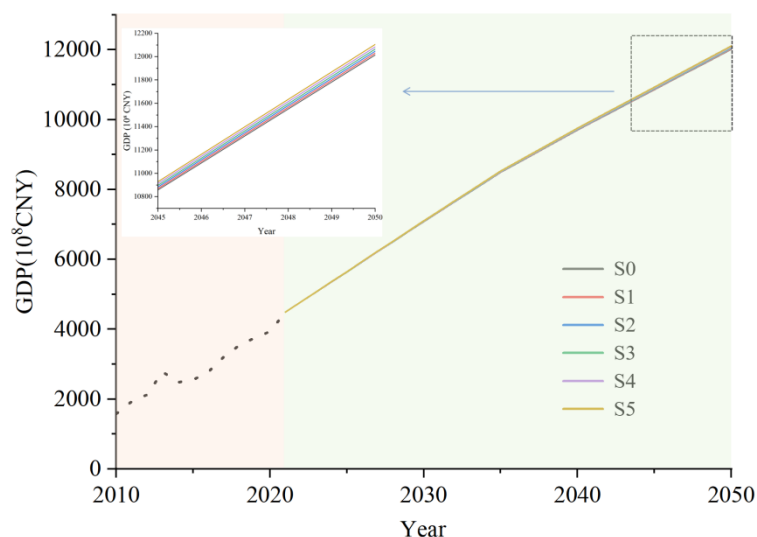


Figure 11. Historical measurements and future simulations of GDP under scenarios S1-S5

To comprehensively analyze the synergistic relationships within the WSES Nexus specifically among runoff, sediment load, and ecological area under different afforestation and grassland restoration policy scenarios in the Yellow River Basin a four-dimensional visualization method was employed for the subsystem data correlation diagram (Figure 12). This diagram simultaneously presents information across four dimensions within the same coordinate system: policy type (represented by different shapes), time series (represented by different colors), forest and grassland area (represented by symbol size), and the hydrological-sediment response (represented by the horizontal and vertical axes). By fitting quadratic polynomial trend lines, the study systematically evaluated the soil and water conservation effects and their economic linkages under different intensities of forest and grassland restoration policies (3%, 6%, 9%, 12%, and 15%) during the period from 2030 to 2050.

The intensity of restoration policies shows a significant positive correlation with ecological benefits. Among them, the 15% restoration policy demonstrates the optimal overall performance: by 2050, its sediment output (23.8 million tons) is approximately 73% lower than that under the 3% policy (88.9 million tons), while being associated with the highest GDP level (12,105.1 billion yuan). This indicates a synergistic gain between ecological protection and economic growth. Trend line analysis further reveals that as policy intensity increases, the runoff sediment relationship curve shifts systematically downward and toward the lower left quadrant. This indicates that more intensive restoration measures can more

effectively reduce the absolute total amount of sediment. However, the implementation costs for higher tier policies (12%, 15%) are relatively high, and the associated decline in runoff is more pronounced, which could potentially impact local water demand in a water-scarce region like Ningxia.

To achieve sustainable basin development, the results suggest prioritizing the implementation of the 9% forest and grassland restoration policy in ecologically sensitive areas and key water conservation zones. In general treatment areas, the 6% policy can be adopted as a balanced solution, while in financially constrained regions, the 9% policy could be implemented in phases. When formulating specific policies, it is also essential to comprehensively consider the water use efficiency across different sectors to optimize the allocation of water resources among competing demands, and to gradually elevate restoration targets. This study provides a quantitative basis for developing differentiated ecological restoration strategies and emphasizes the critical role of Nature-based Solutions (NbS) in coordinating ecological security and regional development.

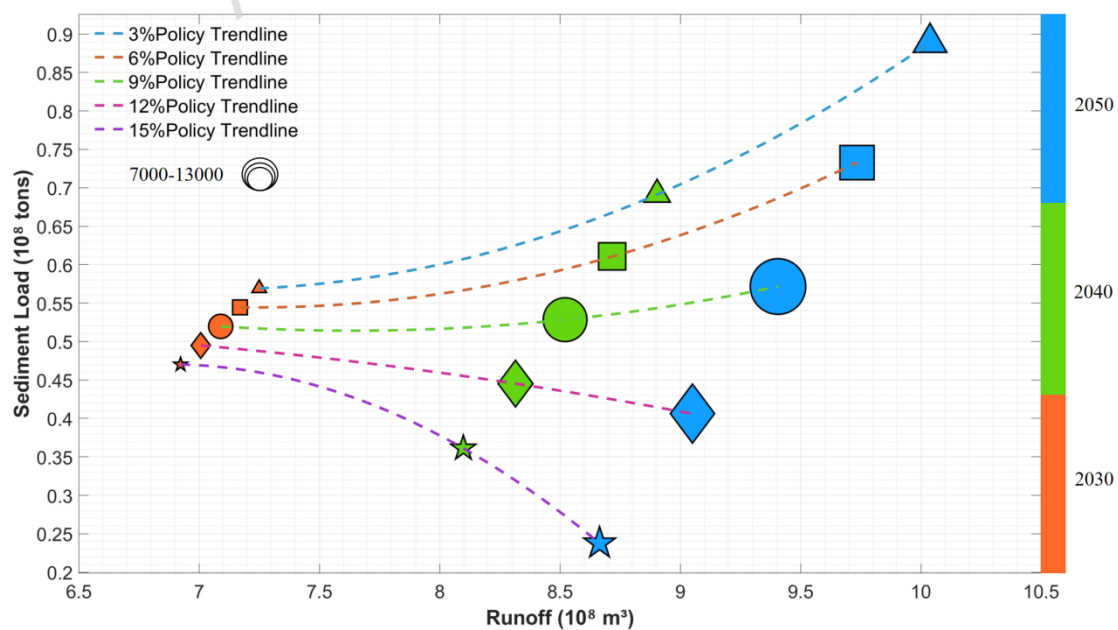


Figure 12. Subsystem data correlation diagram under policy scenarios. A multi-dimensional scatter plot showing the relationships between runoff, sediment load, and GDP under different forest and grassland restoration policies (2030-2050).

Note: Each point's color represents a specific year (see color bar at right) and its shape represents a specific restoration policy (Δ 3%, $\square$ 6%, $\circ$ 9%, $\diamond$ 12%, $\star$ 15%). The point size indicates the GDP level (unit: billion yuan), ranging from 700 to 1300 billion. Larger points represent higher GDP values, with the specific size correspondence shown in the legend at the upper left. The five dashed lines are the quadratic polynomial fitted trend lines for the runoff-sediment relationship within each policy group, reflecting differences in the soil erosion response curves under varying restoration intensities.

4.4 WSES Nexus under Future Climate Change

Impacts

To better investigate the complex processes of competition, feedback, and co-evolution within the system under different climatic futures and socio-economic development pathways, this study further analyzed the impacts of precipitation changes projected by different CMIP6 models on future runoff in the Yellow River Basin. The results are presented as a multi-scenario confidence interval plot (Figure 13). This figure displays the simulated trends and 95% confidence intervals of runoff under different SSP climate scenarios from 2020 to 2050. The analysis indicates that no statistically consistent trend in runoff is observed over the entire study period (e.g., Mann-Kendall test P-values for both SSP1-2.6 and SSP5-8.5 scenarios are greater than 0.050), highlighting the significant uncertainty in future hydrological

projections. Furthermore, this range of uncertainty gradually widens over time.

Across different CMIP6 scenarios, analysis of the entire 41-year period shows that the Mann-Kendall (MK) test statistic ranges from $Z = 0.430$ ($P = 0.670$) under the SSP1-2.6 pathway to $Z = 0.560$ ($P = 0.570$) under the SSP5-8.5 pathway (Table 5). The P-values are substantially greater than the 0.05 significance level, indicating that, over the full temporal scale, the complex fluctuations in runoff do not constitute a statistically significant monotonic trend. A comparison of the scenarios reveals that under high radiative forcing pathways (e.g., SSP3-7.0), runoff exhibits more pronounced variability and wider confidence intervals. This suggests that elevated greenhouse gas emissions would exacerbate the instability and unpredictability of the hydrological response. Conversely, changes under low forcing pathways (e.g., SSP1-2.6, SSP2-4.5) are relatively moderate, with narrower uncertainty ranges. This implies that proactive emission mitigation can offer a more stable and predictable outlook for basin water resource management.

These findings indicate that the choice of climate pathway directly determines the severity and controllability of future water resource risks. Under conditions of expanding uncertainty, a single optimal prediction becomes unreliable. This provides a key scientific basis for the robust management of water-sediment-ecology-economy systems in river basins.

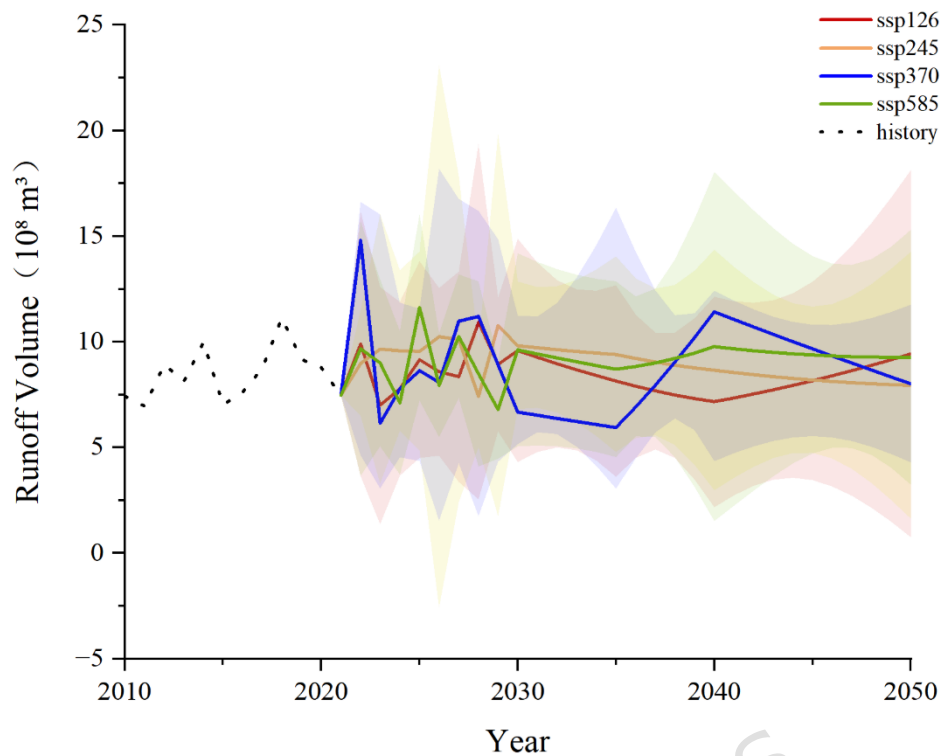


Figure 13. Historical and Simulated Runoff under Different SSPs Climate Scenarios

Table 5. Simulated Runoff Parameter Values under Different SSPs Pathways

Developm ent Pathway	Mean	Std. Dev.	Varianc e	CV (%)	MK Z Statisti c	MK P Statisti c
SSP126	8.509	0.938	0.880	11.020%	0.430	0.670
SSP245	8.756	0.922	0.850	10.530%	0.030	0.980
SSP360	8.797	0.981	0.963	11.150%	-1.500	0.130
SSP585	9.075	0.862	0.743	9.490%	0.560	0.570

Note: The Z-value is a quantitative measure of trend strength in the Mann-Kendall (MK) test. It is derived by standardizing the original trend statistic S , allowing its result to be compared against the standard normal distribution. The P - value represents the probability of observing the current data, or more extreme data,

under the assumption that the null hypothesis is true, i.e., that there are no actual trend and all observed variations are random.

Figure 14 presents the projected range of future sediment load changes in the Yellow River Basin under different Shared Socioeconomic Pathway (SSP) climate scenarios, displayed as a multi scenario confidence interval plot. The figure illustrates the simulated trends and the 95% confidence intervals of sediment load in the Yellow River Basin from 2020 to 2050 under various SSPs scenarios. Simulation results indicate that future sediment load is subject to high uncertainty, with the uncertainty range progressively widening over time. Trend analysis shows that, except for a statistically significant increasing trend detected under the SSP1-2.6 pathway, no consistent monotonic directional change was observed in the other scenarios.

Specifically, under medium to high-forcing scenarios (e.g., SSP2-4.5, SSP3-7.0, and SSP5-8.5), the projected sediment load exhibits greater variability and wider confidence intervals. This suggests that under stronger climatic forcing, extreme hydrological events may exacerbate soil erosion risks, while the nonlinear responses within the precipitation-runoff-sediment transport processes further increase prediction uncertainty. In contrast, the trend under the low-forcing scenario (SSP1-2.6) is relatively moderate with a narrower uncertainty range, indicating that a proactive emission mitigation pathway can help stabilize watershed underlying surface conditions, thereby reducing long-term fluctuations and risks associated with sediment transport. According to the M-K test results across different CMIP6 scenarios over the entire 41-year period (Table 6), a statistically significant increasing trend was detected only under the SSP1-2.6 pathway (Z

= 2.060, $P = 0.013 < 0.050$). Trend analysis confirms a significant upward trend in sediment load throughout the 2010-2050 period under the SSP1-2.6 pathway. No statistically significant monotonic trend was detected for the other pathways based on their M-K statistic values.

This analysis indicates that climate change pathways profoundly influence the uncertainty boundaries of erosion and sediment export processes by regulating precipitation and runoff. Future sediment load does not follow a single trajectory. Therefore, water-sediment regulation and ecological security strategies must fully account for its certainty and its upper and lower bounds. This is crucial for ensuring the robust and sustainable development of the river basin system under varying climatic conditions.

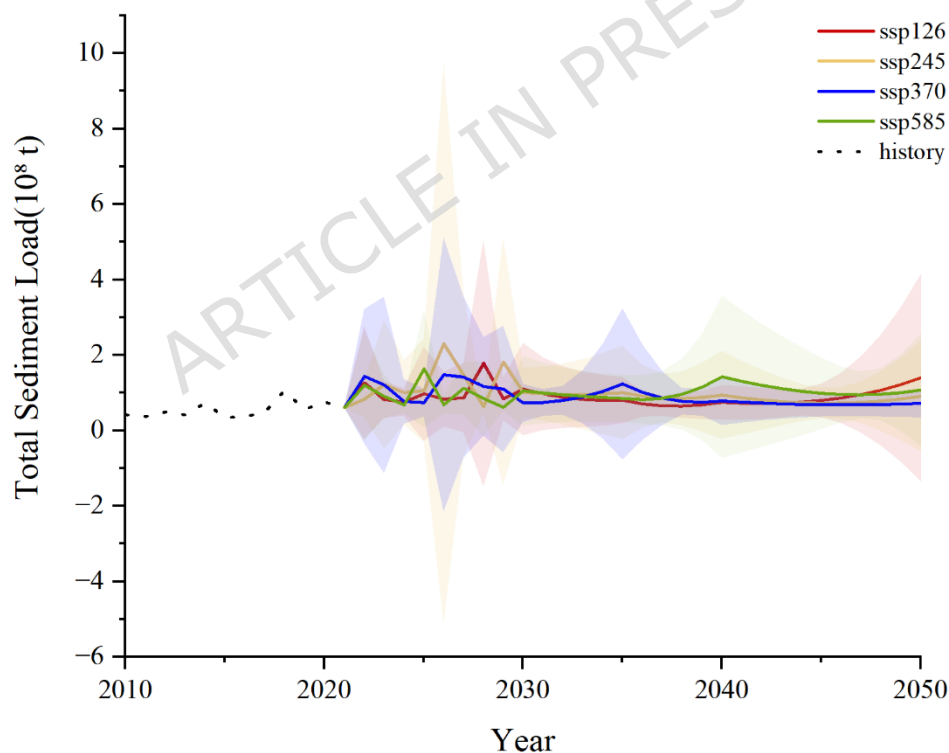


Figure 14. Historical and Simulated Sediment Load under Different SSPs Climate Scenarios

Table 6. Simulated Sediment Load Parameter Values under Different SSPs Pathways

Developm ent Pathway	Mean	Std. Dev.	Varianc e	CV (%)	MK Z Statisti c	MK P Statisti c
SSP126	0.784	0.255	0.065	32.580%	2.060	0.013
SSP245	0.858	0.298	0.089	34.740%	1.110	0.270
SSP360	0.807	0.263	0.069	32.550%	0.460	0.650
SSP585	0.876	0.289	0.083	32.930%	1.840	0.070

To comprehensively investigate the relationships between runoff, water, and sediment under various climate scenarios, this study, based on CMIP6 multi model and multi scenario data, analyzed the response patterns of basin hydrology and sediment to different Shared Socioeconomic Pathway combined with Representative Concentration Pathway (SSPs-RCPs) scenarios from 2030 to 2050. A set of multi-dimensional scatter plots was constructed: Figure 15(a) presents the subsystem data correlation under climate scenarios, Figure 15(b) shows the correlation analysis between runoff and sediment load, and Figure 15(c) depicts the correlation analysis between precipitation and runoff. The results indicate an extremely strong positive correlation between runoff and sediment load ($r = 0.939$). Climate change primarily drives alterations in sediment transport by modifying runoff patterns. Overall, larger data points (representing higher precipitation) generally correspond to higher runoff and higher sediment load, validating the role of precipitation as a key driving force. However, under similar precipitation conditions, sediment yield varies significantly across different scenarios. For instance, at a precipitation level of approximately 18 billion cubic meters, the sediment load under the SSP5-8.5 scenario (77 million tons) is notably lower than that under the SSP1-2.6 scenario (85 million

tons). This suggests that the combined effects of climatic background (e.g., rainfall intensity and frequency) and underlying surface conditions (potentially influenced by climate change) alter sediment yield efficiency.

The results demonstrate that the selection of climate scenarios has a decisive influence on hydrological and sediment processes. Under the SSP5-8.5 (fossil fueled development) scenario, both runoff and sediment load reached their highest levels (averaging 8.990 billion m³ and 80.00 million tons, respectively), exceeding the values for the SSP3-7.0 scenario by 9.6% and 21.2%. This demonstrates the relatively high ecological risks associated with high-emission pathways. A significant linear relationship was observed between precipitation and runoff ($r = 0.820$, indicating a strong correlation), reflects the regulatory role of climate in runoff generation reflecting the regulatory effect of underlying surface changes on runoff generation. None of the scenarios exhibited a consistent linear evolutionary trend from 2030 to 2050. Taking SSP2-4.5 as an example, data points representing high runoff and sediment load in 2030 shifted position by 2040, before moving significantly into a low runoff and sediment load region by 2050. This pattern reflects a phased adjustment in the hydrological response due to changing precipitation patterns under this scenario. Data points for the SSP5-8.5 scenario remained in the high-value range with some fluctuation over time, indicating persistent and relatively stable soil erosion pressure under this high emission pathway. The points for the three sampled years within the SSP3-7.0 scenario clustered within a relatively compact range, exhibiting the lowest coefficients of variation (CV for runoff: 4.0%; CV for sediment load: 5.2%). This suggests a more stable

response from the hydrological-sediment system under this pathway. In contrast, under the SSP1-2.6 scenario, the coefficient of variation of sediment load is as high as 24.9%, with a wider distribution range of data points. In contrast, data points for the SSP1-2.6 and SSP5-8.5 scenarios showed a wider distribution, particularly for sediment load under SSP1-2.6, which had a CV as high as 24.9%. This reveals that, even within a sustainable development framework (SSP1-2.6), the system may still face more complex external climate fluctuations, leading to increased interannual uncertainty in sediment deposition risk.

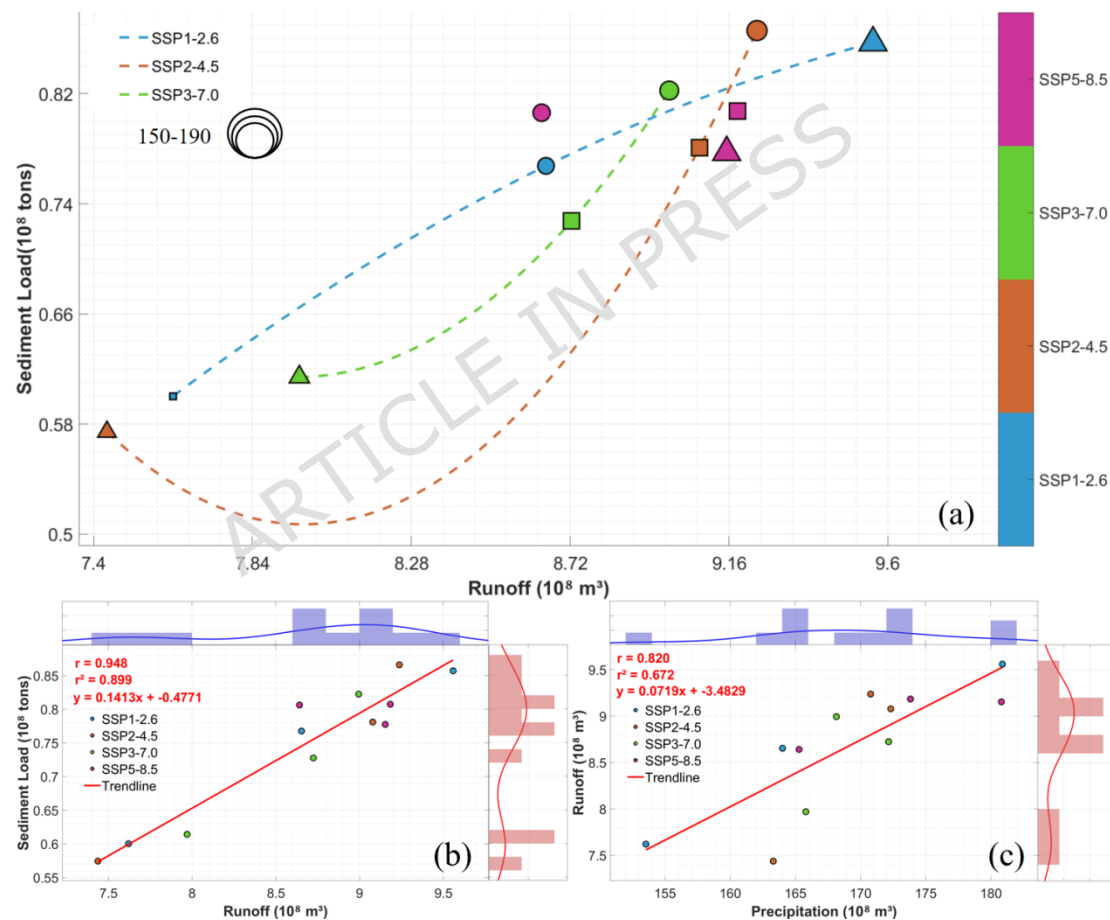


Figure 15. (a) Subsystem data correlation under climate scenarios; (b) Correlation analysis between runoff and sediment load; (c) Correlation analysis between precipitation and runoff.

Note: This multi-dimensional scatter plot illustrates the relationships among runoff, sediment load, and precipitation under

different climate scenarios. The shape of each point indicates the corresponding year: ● 2030, ■ 2040, ▲ 2050. The color of each point represents the climate scenario (as shown in the legend on the right). The size of each point denotes precipitation (unit: 10^8 m^3). The horizontal axis represents runoff (unit: 10^8 m^3), and the vertical axis represents sediment load (unit: 10^8 t).

5. Conclusions

Addressing the systemic challenges arising from the deep coupling and nonlinear feedbacks among water resources, sediment processes, ecological restoration, and socio-economic development under changing environments, this study developed an integrated System Dynamics model for the Ningxia reach of the Yellow River Basin. Compared to existing CHANS-based modeling studies in the Yellow River Basin, this study has the advantages of explicitly coupling sediment processes with hydrological and socio-economic subsystems, as well as integrating ecological restoration policies and SSP-RCP climate pathways into a unified sustainable development framework. The model structure is also tailored to the unique arid-region characteristics and strong human interventions in the Ningxia reach. By embedding both climate uncertainty and human management interventions into a unified WSES Nexus framework, the model captures the long-term co-evolutionary behavior of this strongly human-modified river system. Through scenario-based simulations, the study systematically examined system responses, trade-offs, and vulnerability boundaries under combined policy regulation and climate-socioeconomic pathways, providing a dynamic and quantitative basis for adaptive basin management. The main conclusions of this paper are as follows:

(1) The results of factor screening and sensitivity analysis show that: precipitation has the highest grey relational degree with runoff (0.92), and precipitation is a highly sensitive parameter for both runoff ($S = 1.21$) and sediment load ($S = 1.35$); the grey relational degree between forest and grassland area and sediment load is 0.78, and the growth rate of forest and grassland area exhibits negative sensitivity to sediment load ($S = -0.78$) but has a weaker effect on runoff; the grey relational degree between water use quota and runoff is 0.71, all exceeding the threshold of 0.7, while water use efficiency has limited sensitivity to runoff ($S = -0.34$). The fluctuations of key output variables are all within a reasonable range, constituting the core driving forces of runoff and sediment changes.

(2) Under afforestation and grassland restoration scenarios with varying intensities, a clear nonlinear relationship emerged between ecological benefits and hydrological responses. While higher restoration intensities yielded stronger sediment reduction and ecosystem improvement, they also led to increasingly pronounced runoff reductions. The 15% restoration intensity scenario performs best in terms of ecological and economic benefits, while the 9% restoration intensity provides a balanced and adaptive pathway for regulating the water-sediment relationship.

(3) The study shows that across all SSP-RCP scenarios, runoff and sediment load exhibit a significant positive correlation, and climate and socioeconomic pathways have a substantial impact on future hydrological processes and sediment transport. System volatility and environmental risks are more pronounced under high-emission pathways.

This study constructs a coupled WSES system dynamics model for the Ningxia reach of the Yellow River Basin, revealing the co-evolutionary patterns of regional human-water systems under changing environments. Future research can further incorporate richer social factors (e.g., policy response functions and multi-user agent-based decision models) into the system dynamics model, and couple it with physical process models such as SWAT to deepen the exploration of socio-hydrological coupling mechanisms, thereby enhancing the model's ability to simulate complex socio-hydrological feedbacks and supporting sustainable regional management under changing environmental conditions.

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Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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